

Predictive Maintenance & Fleet RUL Optimization

Executive Engineering & Financial Report (SCIP Framework)

Predictive Analytics & Maintenance Engineering Team

August 22, 2026

Executive Overview

Problem Statement: Turbofan engine failure during operation carries catastrophic safety risks and financial penalties. Traditional fixed-interval maintenance strategies are inefficient, often leading to either unscheduled in-flight groundings or premature replacement of functional parts.

Objective: Predict the Remaining Useful Life (RUL) of individual engine units in real time using multi-sensor thermodynamic telemetry and establish an optimal maintenance action threshold that minimizes total operational expenditure across the fleet.

Key Findings & Actionable Recommendation

- Deploying a LightGBM regression architecture trained on 5-cycle rolling temporal features achieves a cross-validated RMSE of **16.2 cycles**.
- Pure machine learning metrics fail to capture financial exposure; cost-curve simulation reveals an optimal operational intervention threshold at **RUL ≤ 10 cycles**.
- Implementing this threshold reduces total fleet operational cost to **\$543.6k**, successfully eliminating unplanned in-flight failures while maximizing remaining component utility.

1 Situation: Telemetry Drift & Feature Mapping

The operational reliability of turbofan engine fleets depends heavily on continuous telemetry monitoring across key thermodynamic parameters. Using NASA's C-MAPSS FD001 dataset—spanning 100 engine units across 20,631 operational cycles—sensor telemetry tracks progressive mechanical degradation from baseline operation to structural failure.

Through rigorous exploratory data analysis, zero-variance sensor removal, and Variance Inflation Factor (VIF) collinearity checks, key physical indicators of engine health were isolated:

- **Thermal Stress Metrics:** Total Temperature at LPT Outlet (T_{50} / s_4) and HPC Outlet (T_{30} / s_3).
- **Pressure & Aerodynamic Efficiency:** Static Pressure at HPC Outlet (P_{s30} / s_{11}) and Total Pressure (P_{30} / s_7).
- **Rotor Dynamics & Mass Flow:** Corrected Core Speed (NR_c / s_{14}) and Fuel Flow Ratio (ϕ / s_{12}).

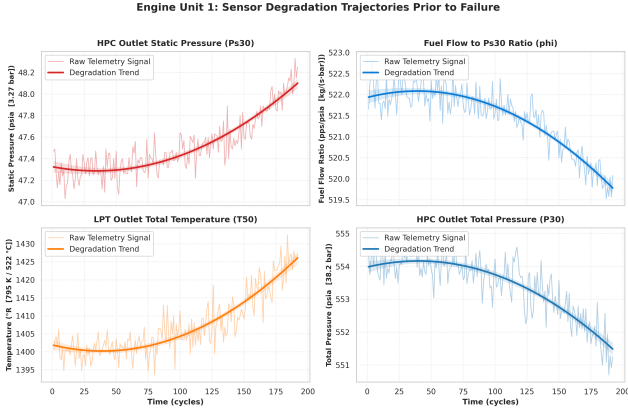


Figure 1: Single-Unit Telemetry Degradation Trajectories Prior to Failure.

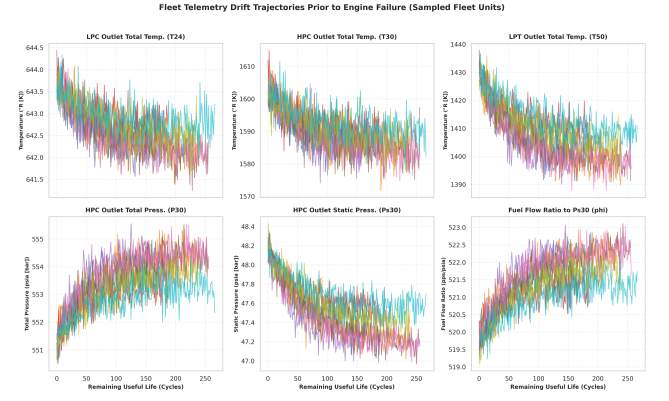


Figure 2: Fleet-Wide Telemetry Drift Trajectories Across Sampled Units.

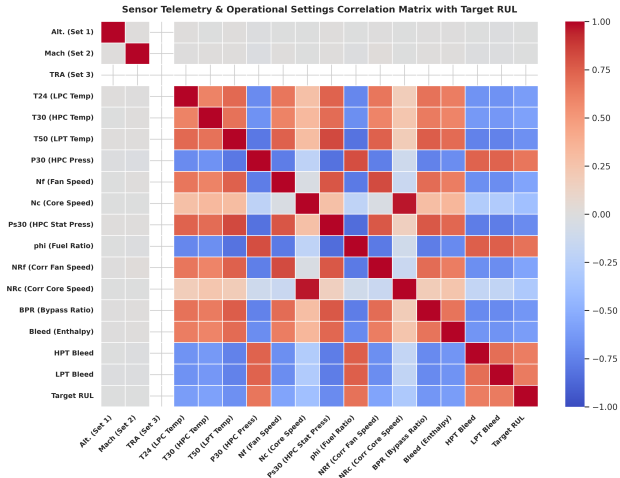


Figure 3: Pearson correlation matrix mapping sensor telemetry drift against target RUL.

2 Constraint: Financial Exposure & Cost Matrix

In real-world deployment, fleet operators face a strict dual-cost operational trade-off:

1. **Catastrophic Unplanned Failure Penalty** ($C_{\text{unplanned}} = \$50,000$): In-flight engine failure or emergency grounding incurs high safety risks, flight cancellations, and emergency labor costs.
2. **Premature Maintenance Waste** ($C_{\text{waste}} = \$50/\text{cycle}$): Pulling an engine out of service prematurely discards unused lifespan across expensive hot-section components.
3. **Scheduled Overhaul Base Cost** ($C_{\text{preventive}} = \$5,000$): Standard fixed cost for planned shop visits and part replacements.

3 Impact: Predictive Tracking & Decision Importance

To address this financial exposure, a piece-wise linear target cap ($\text{RUL} = 125$ cycles) was applied to account for early-life stability. Subsequently, by engineering 5-cycle rolling temporal features, tree-based ensemble models were evaluated via GroupKFold cross-validation on engine unit IDs.

Figure 4 shows the LightGBM model tracking ground-truth RUL decay over an engine's operational lifecycle. Furthermore, Figure 5 displays the top physical indicators driving decision splits, highlighting Corrected Core Speed (NR_c) and Bleed Enthalpy as primary degradation markers.

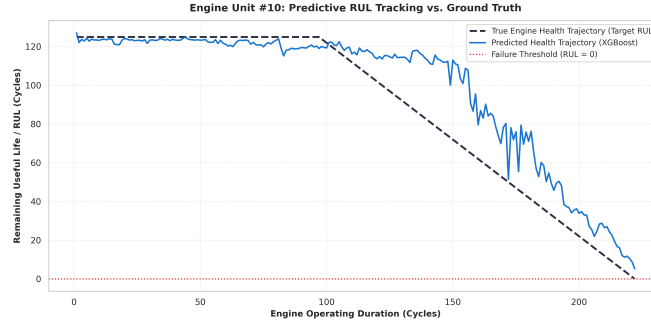


Figure 4: Predicted RUL health trajectory vs. true capped ground truth for Engine Unit #10.

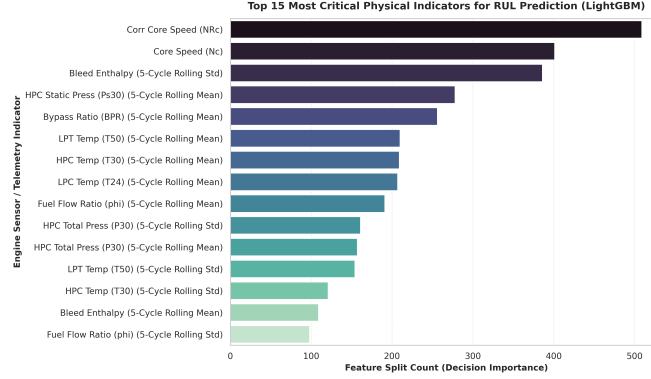


Figure 5: Top 15 physical sensor indicators ranked by LightGBM decision split importance.

4 Path Forward: Actionable Recommendations

Evaluating out-of-fold model predictions against the business cost matrix yields the operational cost curve shown in Figure 6.

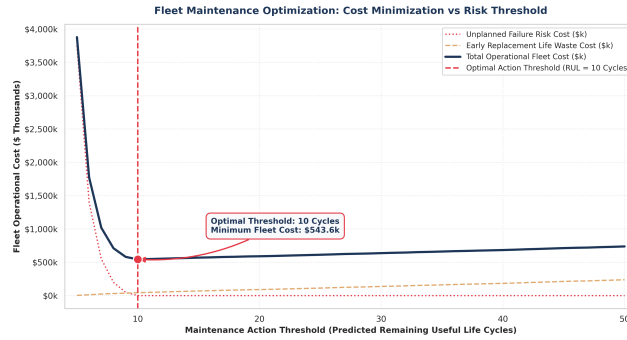


Figure 6: Fleet operational cost minimization balancing unplanned failure penalties against replacement waste.

Strategic Execution Plan

Based on these empirical findings, we recommend executing the following three-step operational roadmap:

1. **Enforce Optimal Action Threshold:** Trigger automated maintenance interventions when predicted remaining life reaches $RUL \leq 10$ cycles, achieving the fleet cost minimum of **\$543.6k**.
2. **Deploy Operational Risk Protocol:**
 - **NORMAL** ($RUL > 25$ cycles): Standard routing; continuous telemetry ingestion.
 - **WARNING** ($10 < RUL \leq 25$ cycles): Flag unit for scheduled servicing during next hub layover.
 - **CRITICAL** ($RUL \leq 10$ cycles): Mandatory grounding and engine overhaul.
3. **Targeted Telemetry Inspections:** Direct shop-level technicians to focus physical teardowns on components associated with NR_c , Bleed Enthalpy, and P_{s30} variance.